

computer vision and image analytics. Table 1 lists these.

OpenCV

OpenCV (<http://opensource.org>) is a high performance library for digital image processing and computer vision, which is free and open source. It is equipped with a large set of functions and algorithms for real-time computer vision and predictive mining. OpenCV was devised and developed by Intel, and the current instances are supported by W. Garage and Itseez. It was developed so that real-time analytics of images and recognition can be done for assorted applications.

The statistical machine learning libraries used by OpenCV are:

- Deep neural networks (DNN)
- Convolutional neural networks (CNN)
- Boosting (meta-algorithm)
- Decision tree learning
- Gradient boosting trees
- Expectation-maximisation algorithm
- K-nearest neighbour algorithm
- Naive Bayes classifier
- Artificial neural networks
- Random forest
- Support vector machine (SVM)

Installation of OpenCV

The installation of OpenCV can be done for different programming languages including Python, Java, C++ and many others. OpenCV provides open connectivity with other platforms and programming languages, so that the algorithms of computer vision can be implemented without any concerns about compatibility or dependencies. It can be installed on different platforms including Android, Windows, Linux, BlackBerry, FreeBSD, OpenBSD, iOS, Maemo or OS X.

OpenCV has open connectivity for installation with different software suites. To integrate OpenCV with Scilab programming, use the following command:

```
--> atomsInstall("scicv")
--> atomsInstall("IPCv")
```

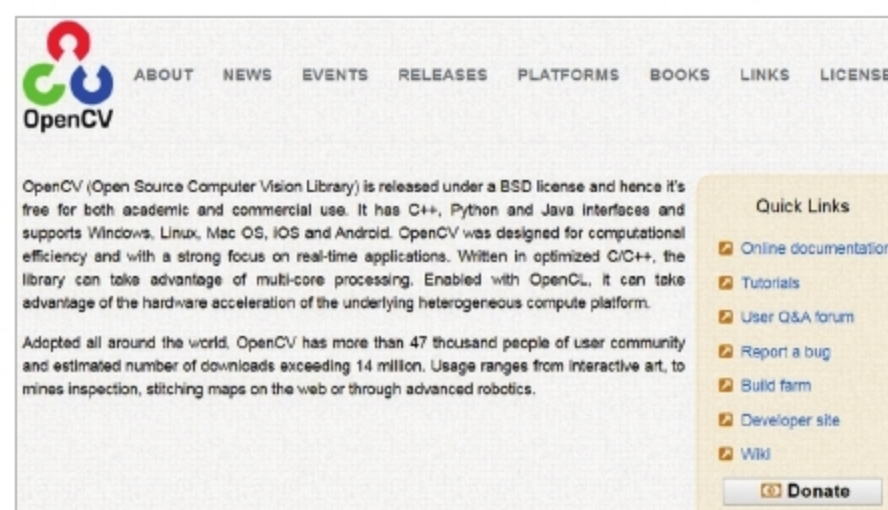


Figure 1: Official portal of OpenCV

Table 1: Tools for computer vision

Tool	URL
OpenCV	http://www.opencv.org/
BoofCV	http://www.boofcv.org/
NASA Vision Workbench	http://ti.arc.nasa.gov/tech/asr/intelligent-robotics/nasa-vision-workbench/
SimpleCV	http://simplecv.org/
Tesseract	https://github.com/tesseract-ocr/tesseract
SLIC Superpixels	http://ivrl.epfl.ch/supplementary_material/RK_SLICSuperpixels/
OpenMVG	https://github.com/openMVG/openMVG
LIBVISO	http://www.cvlibs.net/software/libviso/
VisualSFM	http://homes.cs.washington.edu/~ccwu/vsfm/
MeshLab	http://meshlab.sourceforge.net/
Bundler	http://phototour.cs.washington.edu/bundler/
Vid.stab	https://github.com/georgmartius/vid.stab
ViSP	https://visp.inria.fr/

...or:

```
--> atomsInstall("Source Zip File")
```

To install OpenCV for Python, use the following code:

```
$ pip install opencv-python
$ pip install opencv-contrib-python
```

Real-time image capturing from a Web cam using OpenCV

In traditional implementations, the feature points of the images and computer vision files are recognised on the pre-saved disk images. This approach can be further enhanced using OpenCV, when the real-time video can be marked with the feature points or key points of the image frame in a live running video. The following is the source code of OpenCV that is executed in the Python environment. It can be used to implement real-time recognition of live feature points from a video shot on a Web cam.

```
import numpy as mynp
import cv2
cap = cv2.VideoCapture(0)
while(True):
    ret, frame = cap.read()
    MyGray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    MyGray = mynp.float32(MyGray)
    tdst = cv2.cornerHarris(MyGray,2,3,0.04)
    tdst = cv2.dilate(tdst,None)
    frame[tdst>0.01*tdst.max()]=[0,0,255]
    cv2.imshow('tdst',frame)
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break
cap.release()
cv2.destroyAllWindows()
```